

Hospital Management System Analysis

HOSPITAL



HOSPITAL MANAGEMENT SYSTEM

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INTRODUCTION

The Hospital Management System is a database project developed using MySQL to efficiently manage hospital operations. It stores and organizes information related to patients, doctors, appointments, treatments, and billing in a structured database. The project demonstrates the implementation of SQL concepts such as DDL, DML, Joins, Aggregate Functions, Subqueries, Group By, Window Functions, and CRUD operations to perform real-world hospital data management.

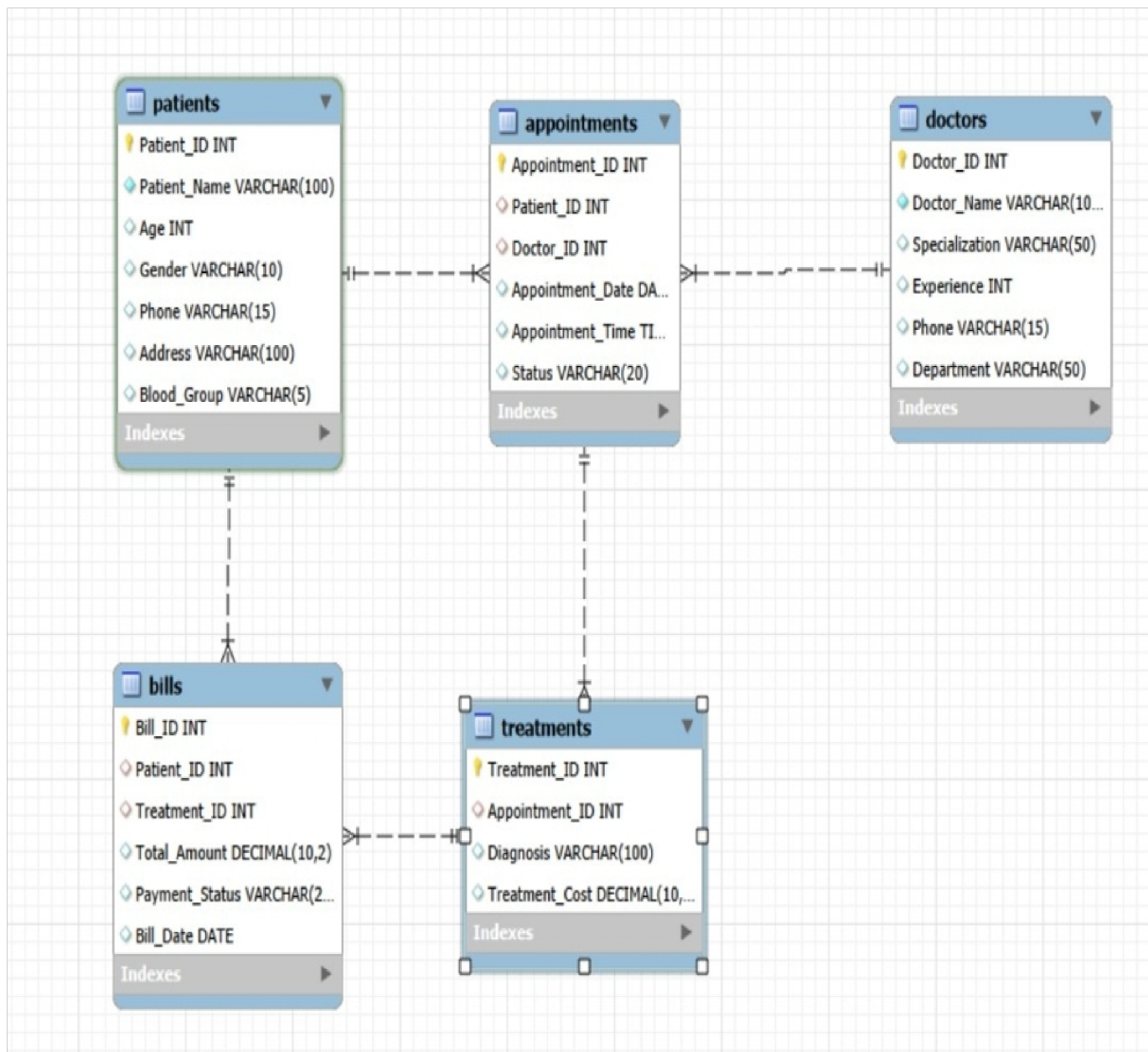
Project Aim

To develop a Hospital Management System using MySQL that efficiently manages hospital records and demonstrates the practical implementation of SQL database concepts for storing, retrieving, updating, and analyzing hospital data.

Project Objectives

- Design a relational database for hospital management.
- Store patient, doctor, appointment, treatment, and billing information.
- Perform CRUD (Create, Read, Update, Delete) operations.
- Retrieve meaningful information using SQL queries.
- Implement Joins, Aggregate Functions, Subqueries, GROUP BY, HAVING, and Window Functions.
- Maintain data accuracy and reduce manual record management.

ER DIAGRAM:

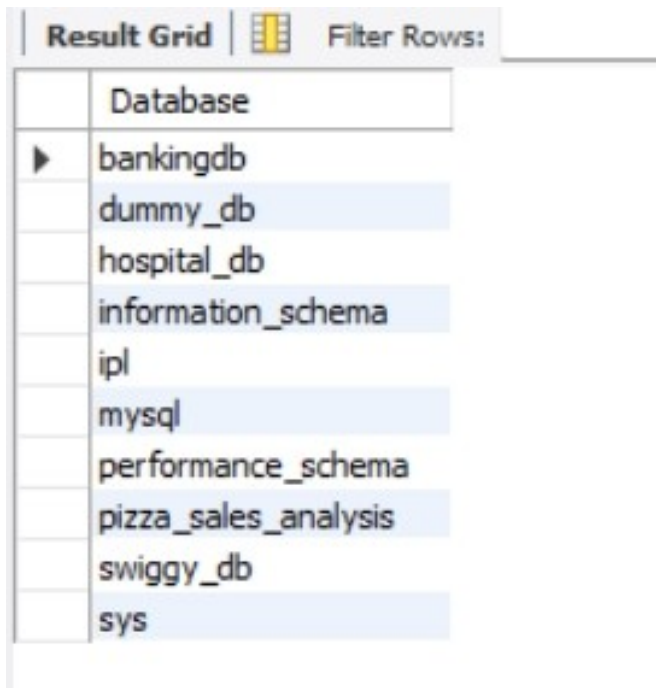


CREATING DATABASE :

CREATE DATABASE Hospital_db;

SHOW DATABASES;

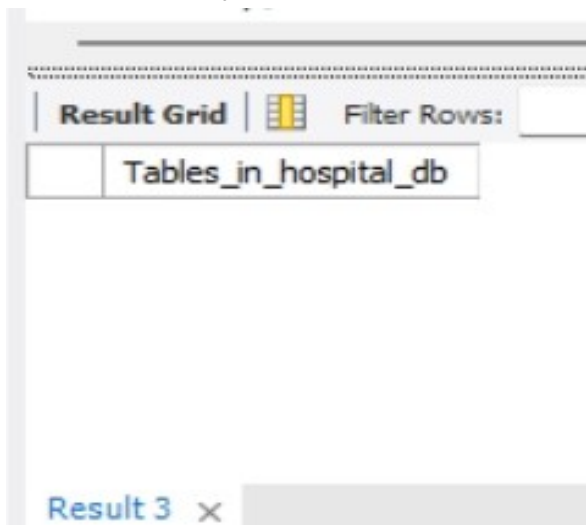
OUTPUT:



	Database
▶	bankingdb
	dummy_db
	hospital_db
	information_schema
	ipl
	mysql
	performance_schema
	pizza_sales_analysis
	swiggy_db
	sys

USE Hospital_db;

SHOW TABLES;



	Tables_in_hospital_db
--	-----------------------

Result 3 ×

OUTPUT-> No tables found (database is empty).

DATA DEFINITION LANGUAGE (DDL) :

CREATING TABLE :

1. Patients Table

```
CREATETABLE Patients (  
    Patient_ID INT PRIMARY KEY,  
    Patient_Name VARCHAR(100) NOT NULL,  
    Age INT,  
    Gender VARCHAR(10),  
    Phone VARCHAR(15) UNIQUE,  
    Address VARCHAR(100),  
    Blood_Group VARCHAR(5)  
);  
DESC Patients;
```

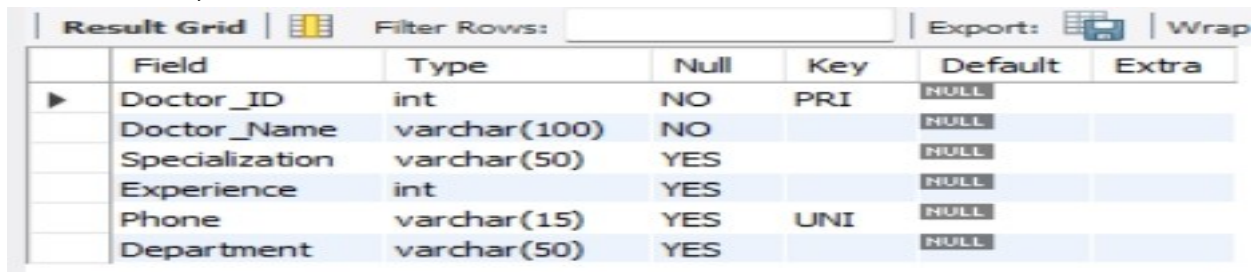
Result Grid Filter Rows: Export: Wrap Cell Content:						
	Field	Type	Null	Key	Default	Extra
▶	Patient_ID	int	NO	PRI	NULL	
	Patient_Name	varchar(100)	NO		NULL	
	Age	int	YES		NULL	
	Gender	varchar(10)	YES		NULL	
	Phone	varchar(15)	YES	UNI	NULL	
	Address	varchar(100)	YES		NULL	
	Blood_Group	varchar(5)	YES		NULL	

OUTPUT-> Patients Table Created Successfully & also table structured displayed successfully.

2.Doctors Table

```
CREATE TABLE Doctors (  
    Doctor_ID INT PRIMARY KEY,  
    Doctor_Name VARCHAR(100) NOT NULL,  
    Specialization VARCHAR(50),  
    Experience INT,  
    Phone VARCHAR(15) UNIQUE,  
    Department VARCHAR(50)  
);
```

DESC Doctors;



The screenshot shows a 'Result Grid' window with a table structure for 'Doctors'. The table has 7 columns: Field, Type, Null, Key, Default, and Extra. The rows are as follows:

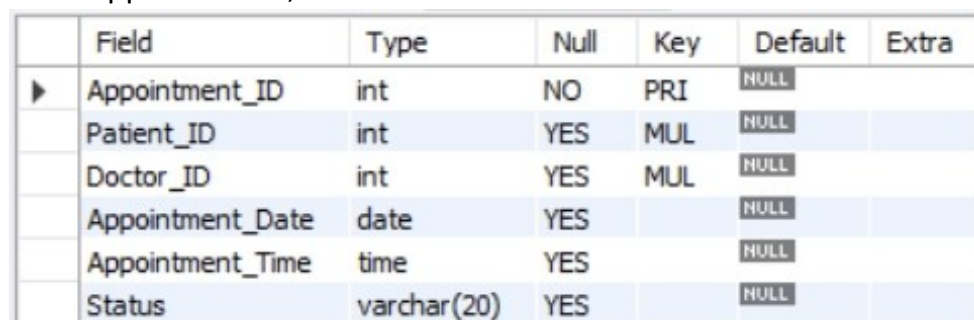
Field	Type	Null	Key	Default	Extra
Doctor_ID	int	NO	PRI	NULL	
Doctor_Name	varchar(100)	NO		NULL	
Specialization	varchar(50)	YES		NULL	
Experience	int	YES		NULL	
Phone	varchar(15)	YES	UNI	NULL	
Department	varchar(50)	YES		NULL	

OUTPUT-> Doctors Table Created Successfully & also table structured displayed successfully.

3. Appointments Table

```
CREATE TABLE Appointments (  
    Appointment_ID INT PRIMARY KEY,  
    Patient_ID INT,  
    Doctor_ID INT,  
    Appointment_Date DATE,  
    Appointment_Time TIME,  
    Status VARCHAR(20),  
    FOREIGN KEY (Patient_ID) REFERENCES Patients(Patient_ID),  
    FOREIGN KEY (Doctor_ID) REFERENCES Doctors(Doctor_ID)  
);
```

DESC Appointments;



The screenshot shows a 'Result Grid' window with a table structure for 'Appointments'. The table has 7 columns: Field, Type, Null, Key, Default, and Extra. The rows are as follows:

Field	Type	Null	Key	Default	Extra
Appointment_ID	int	NO	PRI	NULL	
Patient_ID	int	YES	MUL	NULL	
Doctor_ID	int	YES	MUL	NULL	
Appointment_Date	date	YES		NULL	
Appointment_Time	time	YES		NULL	
Status	varchar(20)	YES		NULL	

OUTPUT->Appointments Table Created Successfully & also table structured displayed successfully.

4.Treatment

```
CREATE TABLE Treatments (  
    Treatment_ID INT PRIMARY KEY,  
    Appointment_ID INT,  
    Diagnosis VARCHAR(100),  
    Treatment_Cost DECIMAL(10,2),  
    FOREIGN KEY (Appointment_ID) REFERENCES Appointments(Appointment_ID)  
);  
DESC Treatments;
```

	Field	Type	Null	Key	Default	Extra
►	Treatment_ID	int	NO	PRI	NULL	
	Appointment_ID	int	YES	MUL	NULL	
	Diagnosis	varchar(100)	YES		NULL	
	Treatment_Cost	decimal(10,2)	YES		NULL	

OUTPUT->Treatment Table Created Successfully & also table structured displayed successfully.

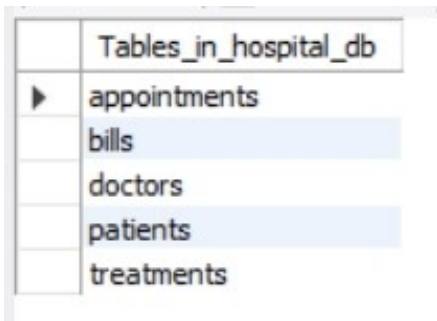
5.Bills Table

```
CREATE TABLE Bills (  
    Bill_ID INT PRIMARY KEY,  
    Patient_ID INT,  
    Treatment_ID INT,  
    Total_Amount DECIMAL(10,2),  
    Payment_Status VARCHAR(20),  
    Bill_Date DATE,  
    FOREIGN KEY (Patient_ID) REFERENCES Patients(Patient_ID),  
    FOREIGN KEY (Treatment_ID) REFERENCES Treatments(Treatment_ID)  
);  
DESC Bills;
```

	Field	Type	Null	Key	Default	Extra
►	Bill_ID	int	NO	PRI	NULL	
	Patient_ID	int	YES	MUL	NULL	
	Treatment_ID	int	YES	MUL	NULL	
	Total_Amount	decimal(10,2)	YES		NULL	
	Payment_Status	varchar(20)	YES		NULL	
	Bill_Date	date	YES		NULL	

OUTPUT->Bill Table Created Successfully & also table structured displayed successfully.

AFTER CREATING ALL TABLE IN Hospital_dbResult:



Tables_in_hospital_db
appointments
bills
doctors
patients
treatments

DATA MANIPULATION LANGUAGE (DML):

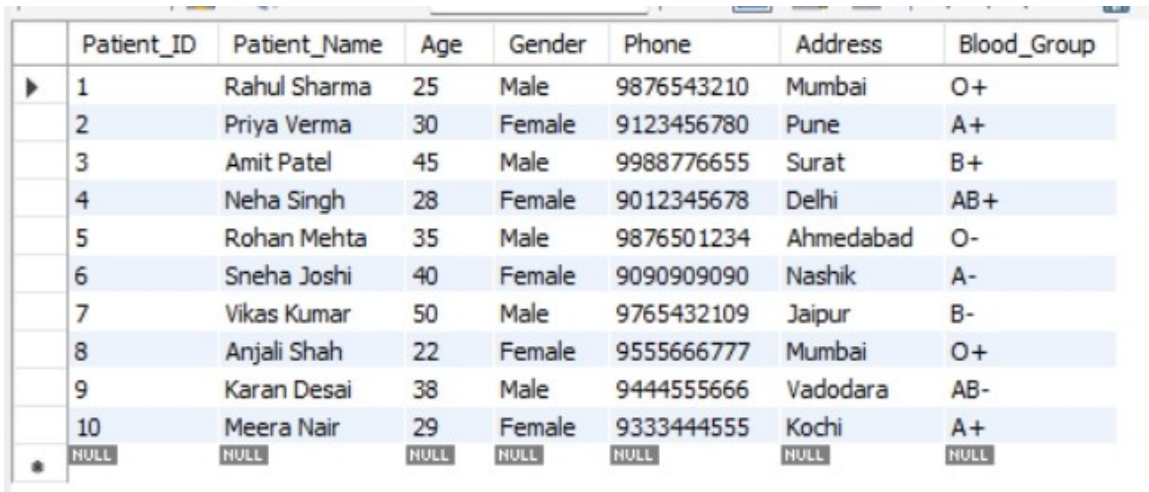
1. Inserting Values into Patient Table

INSERT INTO Patients

VALUES

(1,'Rahul Sharma',25,'Male','9876543210','Mumbai','O+'),
(2,'Priya Verma',30,'Female','9123456780','Pune','A+'),
(3,'Amit Patel',45,'Male','9988776655','Surat','B+'),
(4,'Neha Singh',28,'Female','9012345678','Delhi','AB+'),
(5,'Rohan Mehta',35,'Male','9876501234','Ahmedabad','O-'),
(6,'Sneha Joshi',40,'Female','9090909090','Nashik','A-'),
(7,'Vikas Kumar',50,'Male','9765432109','Jaipur','B-'),
(8,'Anjali Shah',22,'Female','9555666777','Mumbai','O+'),
(9,'Karan Desai',38,'Male','9444555666','Vadodara','AB-'),
(10,'Meera Nair',29,'Female','9333444555','Kochi','A+');

SELECT * FROM Patients;



	Patient_ID	Patient_Name	Age	Gender	Phone	Address	Blood_Group
▶	1	Rahul Sharma	25	Male	9876543210	Mumbai	O+
	2	Priya Verma	30	Female	9123456780	Pune	A+
	3	Amit Patel	45	Male	9988776655	Surat	B+
	4	Neha Singh	28	Female	9012345678	Delhi	AB+
	5	Rohan Mehta	35	Male	9876501234	Ahmedabad	O-
	6	Sneha Joshi	40	Female	9090909090	Nashik	A-
	7	Vikas Kumar	50	Male	9765432109	Jaipur	B-
	8	Anjali Shah	22	Female	9555666777	Mumbai	O+
	9	Karan Desai	38	Male	9444555666	Vadodara	AB-
	10	Meera Nair	29	Female	9333444555	Kochi	A+
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

OUTPUT->10 patient records inserted successfully

2.Inserting Values into Doctors Table

INSERT INTO Doctors

VALUES

(101,'Dr. Amit Sharma','Cardiologist',12,'9876500001','Cardiology'),
(102,'Dr. Neha Verma','Neurologist',8,'9876500002','Neurology'),
(103,'Dr. Raj Patel','Orthopedic',10,'9876500003','Orthopedics'),
(104,'Dr. Priya Singh','Pediatrician',6,'9876500004','Pediatrics'),
(105,'Dr. Karan Mehta','Dermatologist',9,'9876500005','Dermatology'),
(106,'Dr. Sneha Joshi','Gynecologist',11,'9876500006','Gynecology'),
(107,'Dr. Vikas Kumar','ENT Specialist',7,'9876500007','ENT'),
(108,'Dr. Anjali Shah','Ophthalmologist',5,'9876500008','Ophthalmology'),
(109,'Dr. Meera Nair','Psychiatrist',13,'9876500009','Psychiatry'),
(110,'Dr. Rahul Desai','General Physician',15,'9876500010','General Medicine');

SELECT * FROM Doctors;

	Doctor_ID	Doctor_Name	Specialization	Experience	Phone	Department
▶	101	Dr. Amit Sharma	Cardiologist	12	9876500001	Cardiology
	102	Dr. Neha Verma	Neurologist	8	9876500002	Neurology
	103	Dr. Raj Patel	Orthopedic	10	9876500003	Orthopedics
	104	Dr. Priya Singh	Pediatrician	6	9876500004	Pediatrics
	105	Dr. Karan Mehta	Dermatologist	9	9876500005	Dermatology
	106	Dr. Sneha Joshi	Gynecologist	11	9876500006	Gynecology
	107	Dr. Vikas Kumar	ENT Specialist	7	9876500007	ENT
	108	Dr. Anjali Shah	Ophthalmologist	5	9876500008	Ophthalmology
	109	Dr. Meera Nair	Psychiatrist	13	9876500009	Psychiatry
	110	Dr. Rahul Desai	General Physician	15	9876500010	General Medicine
★	NULL	NULL	NULL	NULL	NULL	NULL

OUTPUT->10 patient records inserted successfully

3.Inserting Values into Appointments Table

INSERT INTO Appointments

VALUES

```
(1001,1,101,'2025-07-20','10:00:00','Completed'),  
(1002,2,102,'2025-07-20','11:00:00','Completed'),  
(1003,3,103,'2025-07-21','09:30:00','Pending'),  
(1004,4,104,'2025-07-21','12:00:00','Completed'),  
(1005,5,105,'2025-07-22','10:30:00','Cancelled'),  
(1006,6,106,'2025-07-22','02:00:00','Completed'),  
(1007,7,107,'2025-07-23','11:30:00','Pending'),  
(1008,8,108,'2025-07-23','03:00:00','Completed'),  
(1009,9,109,'2025-07-24','09:00:00','Completed'),  
(1010,10,110,'2025-07-24','04:00:00','Pending');
```

SELECT * FROM Appointments;

	Appointment_ID	Patient_ID	Doctor_ID	Appointment_Date	Appointment_Time	Status
▶	1001	1	101	2025-07-20	10:00:00	Completed
	1002	2	102	2025-07-20	11:00:00	Completed
	1003	3	103	2025-07-21	09:30:00	Pending
	1004	4	104	2025-07-21	12:00:00	Completed
	1005	5	105	2025-07-22	10:30:00	Cancelled
	1006	6	106	2025-07-22	02:00:00	Completed
	1007	7	107	2025-07-23	11:30:00	Pending
	1008	8	108	2025-07-23	03:00:00	Completed
	1009	9	109	2025-07-24	09:00:00	Completed
	1010	10	110	2025-07-24	04:00:00	Pending
•	NULL	NULL	NULL	NULL	NULL	NULL

OUTPUT->10 patient records inserted successfully

4.Inserting Values into Treatement Table

INSERT INTO Treatments

VALUES

(201,1001,'Fever',1500.00),
(202,1002,'Migraine',2500.00),
(203,1003,'Fracture',8000.00),
(204,1004,'Cold & Cough',1200.00),
(205,1005,'Skin Allergy',1800.00),
(206,1006,'Pregnancy Checkup',3000.00),
(207,1007,'Ear Infection',2200.00),
(208,1008,'Eye Checkup',1700.00),
(209,1009,'Depression',4000.00),
(210,1010,'General Checkup',1000.00);

SELECT * FROM Treatments;

	Treatment_ID	Appointment_ID	Diagnosis	Treatment_Cost
▶	201	1001	Fever	1500.00
	202	1002	Migraine	2500.00
	203	1003	Fracture	8000.00
	204	1004	Cold & Cough	1200.00
	205	1005	Skin Allergy	1800.00
	206	1006	Pregnancy Checkup	3000.00
	207	1007	Ear Infection	2200.00
	208	1008	Eye Checkup	1700.00
	209	1009	Depression	4000.00
	210	1010	General Checkup	1000.00
●	NULL	NULL	NULL	NULL

OUTPUT->10 patient records inserted successfully

5.Inserting Values into Bills Table

INSERT INTO Bills

VALUES

```
(301,1,201,1500.00,'Paid','2025-07-20'),  
(302,2,202,2500.00,'Paid','2025-07-20'),  
(303,3,203,8000.00,'Pending','2025-07-21'),  
(304,4,204,1200.00,'Paid','2025-07-21'),  
(305,5,205,1800.00,'Cancelled','2025-07-22'),  
(306,6,206,3000.00,'Paid','2025-07-22'),  
(307,7,207,2200.00,'Pending','2025-07-23'),  
(308,8,208,1700.00,'Paid','2025-07-23'),  
(309,9,209,4000.00,'Paid','2025-07-24'),  
(310,10,210,1000.00,'Paid','2025-07-24');
```

SELECT * FROM Bills;

	Bill_ID	Patient_ID	Treatment_ID	Total_Amount	Payment_Status	Bill_Date
▶	301	1	201	1500.00	Paid	2025-07-20
	302	2	202	2500.00	Paid	2025-07-20
	303	3	203	8000.00	Pending	2025-07-21
	304	4	204	1200.00	Paid	2025-07-21
	305	5	205	1800.00	Cancelled	2025-07-22
	306	6	206	3000.00	Paid	2025-07-22
	307	7	207	2200.00	Pending	2025-07-23
	308	8	208	1700.00	Paid	2025-07-23
	309	9	209	4000.00	Paid	2025-07-24
	310	10	210	1000.00	Paid	2025-07-24
★	NULL	NULL	NULL	NULL	NULL	NULL

OUTPUT->10 bills records inserted successfully

BASIC QUESTIONS

1. Basic Queries

1. Display the details of patients whose age is above 30 years.

```
SELECT *  
FROM Patients  
WHERE Age > 30;
```

	Patient_ID	Patient_Name	Age	Gender	Phone	Address	Blood_Group
▶	3	Amit Patel	45	Male	9988776655	Surat	B+
	5	Rohan Mehta	35	Male	9876501234	Ahmedabad	O-
	6	Sneha Joshi	40	Female	9090909090	Nashik	A-
	7	Vikas Kumar	50	Male	9765432109	Jaipur	B-
	9	Karan Desai	38	Male	9444555666	Vadodara	AB-
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. Display the names and departments of doctors with more than 8 years of experience.

```
SELECT Doctor_Name, Department, Experience  
FROM Doctors  
WHERE Experience > 8;
```

	Doctor_Name	Department	Experience
▶	Dr. Amit Sharma	Cardiology	12
	Dr. Raj Patel	Orthopedics	10
	Dr. Karan Mehta	Dermatology	9
	Dr. Sneha Joshi	Gynecology	11
	Dr. Meera Nair	Psychiatry	13
	Dr. Rahul Desai	General Medicine	15

3. Display all appointments that are scheduled as 'Pending'.

```
SELECT *  
FROM Appointments  
WHERE Status = 'Pending';
```

	Appointment_ID	Patient_ID	Doctor_ID	Appointment_Date	Appointment_Time	Status
▶	1003	3	103	2025-07-21	09:30:00	Pending
	1007	7	107	2025-07-23	11:30:00	Pending
	1010	10	110	2025-07-24	04:00:00	Pending
*	NULL	NULL	NULL	NULL	NULL	NULL

4. Display the top 5 highest treatment costs.

```
SELECT *  
FROM Treatments  
ORDER BY Treatment_Cost DESC  
LIMIT 5;
```

	Treatment_ID	Appointment_ID	Diagnosis	Treatment_Cost
▶	203	1003	Fracture	8000.00
	209	1009	Depression	4000.00
	206	1006	Pregnancy Checkup	3000.00
	202	1002	Migraine	2500.00
	207	1007	Ear Infection	2200.00
*	NULL	NULL	NULL	NULL

5. Display the details of patients belonging to the 'O+' blood group.

```
SELECT *  
FROM Patients  
WHERE Blood_Group = 'O+';
```

	Patient_ID	Patient_Name	Age	Gender	Phone	Address	Blood_Group
▶	1	Rahul Sharma	25	Male	9876543210	Mumbai	O+
	8	Anjali Shah	22	Female	9555666777	Mumbai	O+
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. Aggregate Functions

6. Find the total number of appointments.

```
SELECT COUNT(*) AS Total_Appointments  
FROM Appointments;
```

	Total_Appointments
▶	10

7. Find the total revenue collected from paid bills.

```
SELECT SUM(Total_Amount) AS Total_Revenue
FROM Bills
WHERE Payment_Status = 'Paid';
```

Total_Revenue
14900.00

8. Find the average treatment cost.

```
SELECT AVG(Treatment_Cost) AS Average_Treatment_Cost
FROM Treatments;
```

Average_Treatment_Cost
2690.000000

9. Find the maximum and minimum bill amount.

```
SELECT MAX(Total_Amount) AS Maximum_Bill,
       MIN(Total_Amount) AS Minimum_Bill
FROM Bills;
```

Highest_Bill	Lowest_Bill
8000.00	1000.00

3. GROUP BY & HAVING

10. Count the number of doctors in each department.

```
SELECT Department,
       COUNT(*) AS Total_Doctors
FROM Doctors
GROUP BY Department;
```

	Appointment_ID	Patient_ID	Doctor_ID	Appointment_Date	Appointment_Time	Status
▶	1001	1	101	2025-07-20	10:00:00	Completed
	1002	2	102	2025-07-20	11:00:00	Completed
	1003	3	103	2025-07-21	09:30:00	Pending
	1004	4	104	2025-07-21	12:00:00	Completed
	1005	5	105	2025-07-22	10:30:00	Cancelled
	1006	6	106	2025-07-22	02:00:00	Completed
	1007	7	107	2025-07-23	11:30:00	Pending
	1008	8	108	2025-07-23	03:00:00	Completed
	1009	9	109	2025-07-24	09:00:00	Completed
	1010	10	110	2025-07-24	04:00:00	Pending
*	NULL	NULL	NULL	NULL	NULL	NULL

11. Calculate the total revenue generated by each payment status.

```
SELECT Payment_Status,  
       SUM(Total_Amount) AS Total_Revenue  
FROM Bills  
GROUP BY Payment_Status;
```

	Treatment_ID	Appointment_ID	Diagnosis	Treatment_Cost
▶	201	1001	Fever	1500.00
	202	1002	Migraine	2500.00
	203	1003	Fracture	8000.00
	204	1004	Cold & Cough	1200.00
	205	1005	Skin Allergy	1800.00
	206	1006	Pregnancy Checkup	3000.00
	207	1007	Ear Infection	2200.00
	208	1008	Eye Checkup	1700.00
	209	1009	Depression	4000.00
	210	1010	General Checkup	1000.00
*	NULL	NULL	NULL	NULL

12. Display payment statuses having total revenue greater than 3000.

```
SELECT Payment_Status,  
       SUM(Total_Amount) AS Total_Revenue  
FROM Bills  
GROUP BY Payment_Status  
HAVING SUM(Total_Amount) > 3000;
```

	Bill_ID	Patient_ID	Treatment_ID	Total_Amount	Payment_Status	Bill_Date
▶	301	1	201	1500.00	Paid	2025-07-20
	302	2	202	2500.00	Paid	2025-07-20
	303	3	203	8000.00	Pending	2025-07-21
	304	4	204	1200.00	Paid	2025-07-21
	305	5	205	1800.00	Cancelled	2025-07-22
	306	6	206	3000.00	Paid	2025-07-22
	307	7	207	2200.00	Pending	2025-07-23
	308	8	208	1700.00	Paid	2025-07-23
	309	9	209	4000.00	Paid	2025-07-24
	310	10	210	1000.00	Paid	2025-07-24
*	NULL	NULL	NULL	NULL	NULL	NULL

4. Joins

13. Display all patients, even if they have no appointment.

```
SELECT P.Patient_Name, A.Appointment_Date  
FROM Patients P  
LEFT JOIN Appointments A  
ON P.Patient_ID = A.Patient_ID;
```

	Patient_Name	Appointment_Date
▶	Rahul Sharma	2025-07-20
	Priya Verma	2025-07-20
	Amit Patel	2025-07-21
	Neha Singh	2025-07-21
	Rohan Mehta	2025-07-22
	Sneha Joshi	2025-07-22
	Vikas Kumar	2025-07-23
	Anjali Shah	2025-07-23
	Karan Desai	2025-07-24
	Meera Nair	2025-07-24
	Arjun Patil	NULL

14. Display all doctors, even if they have no appointments.

```
SELECT D.Doctor_Name, A.Appointment_Date  
FROM Appointments A  
RIGHT JOIN Doctors D  
ON A.Doctor_ID = D.Doctor_ID;
```

	Doctor_Name	Appointment_Date
▶	Dr. Amit Sharma	2025-07-20
	Dr. Neha Verma	2025-07-20
	Dr. Raj Patel	2025-07-21
	Dr. Priya Singh	2025-07-21
	Dr. Karan Mehta	2025-07-22
	Dr. Sneha Joshi	2025-07-22
	Dr. Vikas Kumar	2025-07-23
	Dr. Anjali Shah	2025-07-23
	Dr. Meera Nair	2025-07-24
	Dr. Rahul Desai	2025-07-24

15. Display patient name, doctor name, diagnosis, and bill amount.

```
SELECT P.Patient_Name,  
       D.Doctor_Name,  
       T.Diagnosis,  
       B.Total_Amount  
FROM Bills B  
JOIN Treatments T  
ON B.Treatment_ID = T.Treatment_ID  
JOIN Appointments A  
ON T.Appointment_ID = A.Appointment_ID  
JOIN Patients P  
ON A.Patient_ID = P.Patient_ID  
JOIN Doctors D  
ON A.Doctor_ID = D.Doctor_ID;
```

	Patient_Name	Doctor_Name	Diagnosis	Total_Amount
▶	Rahul Sharma	Dr. Amit Sharma	Fever	1500.00
	Priya Verma	Dr. Neha Verma	Migraine	2500.00
	Amit Patel	Dr. Raj Patel	Fracture	8000.00
	Neha Singh	Dr. Priya Singh	Cold & Cough	1200.00
	Rohan Mehta	Dr. Karan Mehta	Skin Allergy	1800.00
	Sneha Joshi	Dr. Sneha Joshi	Pregnancy Checkup	3000.00
	Vikas Kumar	Dr. Vikas Kumar	Ear Infection	2200.00
	Anjali Shah	Dr. Anjali Shah	Eye Checkup	1700.00
	Karan Desai	Dr. Meera Nair	Depression	4000.00
	Meera Nair	Dr. Rahul Desai	General Checkup	1000.00

5.Subqueries

16.Display patients whose age is greater than the average age.

```
SELECT *  
FROM Patients  
WHERE Age >  
(  
    SELECT AVG(Age)  
    FROM Patients  
);
```

	Patient_ID	Patient_Name	Age	Gender	Phone	Address	Blood_Group
▶	3	Amit Patel	45	Male	9988776655	Surat	B+
	5	Rohan Mehta	35	Male	9876501234	Ahmedabad	O-
	6	Sneha Joshi	40	Female	9090909090	Nashik	A-
	7	Vikas Kumar	50	Male	9765432109	Jaipur	B-
	9	Karan Desai	38	Male	9444555666	Vadodara	AB-
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

17. Find the doctor having the highest experience.

```
SELECT *  
FROM Doctors  
WHERE Experience =  
(  
    SELECT MAX(Experience)  
    FROM Doctors  
);
```

	Doctor_ID	Doctor_Name	Specialization	Experience	Phone	Department
▶	110	Dr. Rahul Desai	General Physician	15	9876500010	General Medicine
*	NULL	NULL	NULL	NULL	NULL	NULL

6. Window Functions

18. Rank doctors based on their experience.

```
SELECT Doctor_Name,  
       Experience,  
       RANK() OVER(ORDER BY Experience DESC) AS Doctor_Rank  
FROM Doctors;
```

	Doctor_Name	Experience	Doctor_Rank
▶	Dr. Rahul Desai	15	1
	Dr. Meera Nair	13	2
	Dr. Amit Sharma	12	3
	Dr. Sneha Joshi	11	4
	Dr. Raj Patel	10	5
	Dr. Karan Mehta	9	6
	Dr. Neha Verma	8	7
	Dr. Vikas Kumar	7	8
	Dr. Priya Singh	6	9
	Dr. Anjali Shah	5	10

19. Assign row numbers to appointments based on appointment date.

```
SELECT Appointment_ID,  
       Appointment_Date,  
       ROW_NUMBER() OVER(ORDER BY Appointment_Date) AS Row_Number  
FROM Appointments;
```

	Appointment_ID	Appointment_Date	Row__Number
▶	1001	2025-07-20	1
	1002	2025-07-20	2
	1003	2025-07-21	3
	1004	2025-07-21	4
	1005	2025-07-22	5
	1006	2025-07-22	6
	1007	2025-07-23	7
	1008	2025-07-23	8
	1009	2025-07-24	9
	1010	2025-07-24	10

20. Find the doctor with the second highest experience.

```
SELECT Doctor_Name,  
experience  
FROM(  
SELECT Doctor_Name,  
Experience,  
DENSE_RANK() OVER(ORDER BY Experience DESC) AS Rank_  
FROM Doctors  
) d  
WHERE Rank_ = 2
```

	Doctor_Name	experience
▶	Dr. Meera Nair	13

7.DML Operations

21. Update the status of all pending appointments to 'Completed'.

```
UPDATE Appointments  
SET Status = 'Completed'  
WHERE Status = 'Pending';
```

	Appointment_ID	Patient_ID	Doctor_ID	Appointment_Date	Appointment_Time	Status
▶	1001	1	101	2025-07-20	10:00:00	Completed
	1002	2	102	2025-07-20	11:00:00	Completed
	1003	3	103	2025-07-21	09:30:00	Completed
	1004	4	104	2025-07-21	12:00:00	Completed
	1005	5	105	2025-07-22	10:30:00	Cancelled
	1006	6	106	2025-07-22	02:00:00	Completed
	1007	7	107	2025-07-23	11:30:00	Completed
	1008	8	108	2025-07-23	03:00:00	Completed
	1009	9	109	2025-07-24	09:00:00	Completed
	1010	10	110	2025-07-24	04:00:00	Completed
*	NULL	NULL	NULL	NULL	NULL	NULL

22. Insert a new patient record.

INSERT INTO Patients

VALUES

(11,'Arjun Patil',27,'Male','9000011111','Pune','B+');

	Patient_ID	Patient_Name	Age	Gender	Phone	Address	Blood_Group
▶	1	Rahul Sharma	25	Male	9876543210	Mumbai	O+
	2	Priya Verma	30	Female	9123456780	Pune	A+
	3	Amit Patel	45	Male	9988776655	Surat	B+
	4	Neha Singh	28	Female	9012345678	Delhi	AB+
	5	Rohan Mehta	35	Male	9876501234	Ahmedabad	O-
	6	Sneha Joshi	40	Female	9090909090	Nashik	A-
	7	Vikas Kumar	50	Male	9765432109	Jaipur	B-
	8	Anjali Shah	22	Female	9555666777	Mumbai	O+
	9	Karan Desai	38	Male	9444555666	Vadodara	AB-
	10	Meera Nair	29	Female	9333444555	Kochi	A+
	11	Arjun Patil	27	Male	9000011111	Pune	B+
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

8.VIEW

23.Create a view named Patient_Doctor_View to display the patient name, doctor name, and appointment date.

CREATE VIEW Patient_Doctor_View AS

SELECT P.Patient_Name,

D.Doctor_Name,

A.Appointment_Date

FROM Appointments A

JOIN Patients P

ON A.Patient_ID = P.Patient_ID

JOIN Doctors D

ON A.Doctor_ID = D.Doctor_ID;

SELECT * FROM Patient_Doctor_View;

	Patient_Name	Doctor_Name	Appointment_Date
▶	Rahul Sharma	Dr. Amit Sharma	2025-07-20
	Priya Verma	Dr. Neha Verma	2025-07-20
	Amit Patel	Dr. Raj Patel	2025-07-21
	Neha Singh	Dr. Priya Singh	2025-07-21
	Rohan Mehta	Dr. Karan Mehta	2025-07-22
	Sneha Joshi	Dr. Sneha Joshi	2025-07-22
	Vikas Kumar	Dr. Vikas Kumar	2025-07-23
	Anjali Shah	Dr. Anjali Shah	2025-07-23
	Karan Desai	Dr. Meera Nair	2025-07-24
	Meera Nair	Dr. Rahul Desai	2025-07-24

24. Using the Patient_Doctor_View, display all appointments scheduled on '2025-07-20'.

```
SELECT *  
FROM Patient_Doctor_View  
WHERE Appointment_Date = '2025-07-20';
```

	Patient_Name	Doctor_Name	Appointment_Date
▶	Rahul Sharma	Dr. Amit Sharma	2025-07-20
	Priya Verma	Dr. Neha Verma	2025-07-20

Conclusion

The **Hospital_DB** project is a SQL-based relational database designed to efficiently manage hospital operations by organizing data related to **Patients, Doctors, Appointments, Treatments, and Billing**. The project applies key SQL concepts such as **Database Design, Primary Keys, Foreign Keys, Constraints, CRUD Operations, Joins, Subqueries, Aggregate Functions, and Views** to ensure accurate data storage, maintain data integrity, and enable efficient data retrieval. Overall, this project demonstrates the practical application of SQL in building a **structured, reliable, and real-world database solution** for healthcare management.